



FEASIBILITY STUDY REPORT

Intelligent System for Pakistan Sign Language Learning and Translation

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RECOMMENDATIONS
(to be filled in by requestor)

The document was prepared by:

[Name]

DATE

[Designation]

[KCDI]

I hereby verify that the content of the document is correct and meets the requirement of my office:

Signature and Stamp

DATE

Project Director

Based on our authority and judgment,

- ☐ ITD Management recommends the FSR **WITHOUT AMENDMENTS** and shall be bring up to UTICTEC Meeting (FSR Meeting)
- ☐ ITD Management recommends the FSR **WITH AMENDMENTS** and shall be bring up to UTICTEC Meeting (amendments as per tabled in the Minute Meeting FSR)
- ☐ The project is to be reviewed again in the next FSR Meeting No.....
- ☐ The project is not recommended.

Signature and Stamp

DATE

Director, ITD

DOCUMENTATION INFORMATION (to be filled in by ITPMO)

Assigned Feasibility Study Report ID : _____

1.0 GENERAL INFORMATION

1.1 Purpose

The purpose of this Feasibility Study is to evaluate the technical, operational, and financial viability of developing an AI-powered mobile application for Pakistan Sign Language (PSL) translation and learning. The system aims to provide real-time PSL-to-text/speech translation, structured PSL lessons, and assessment features to improve communication between hearing and non-hearing communities.

1.2 Scope

1.2.1 In Scope

- Development of a mobile application using **Flutter** (frontend), **Python** (AI models), and **Firestore** (database).
- Creation of a **PSL dictionary** with categorized signs.
- Implementation of **translation module** (PSL ↔ text/speech).
- Structured **learning and assessment modules**.
- Gamification features: leaderboards, streaks, and rewards.
- User profile management, login/signup, and settings.

1.2.2 Out Scope

- Support for sign languages other than PSL (e.g., ASL, BSL).
- Advanced speech synthesis or 3D avatar generation beyond basic PSL animations/videos.
- Large-scale commercial deployment (focus is on prototype/research system).

1.3 Project Overview

Process Owner	FYP Team
Proposed Project Name	Intelligent System for Pakistan Sign Language Learning and Translation
Proposed Project Abbreviation	PSL-TLS (Pakistan Sign Language - Translator & Learning System)

1.4 Project References

- Zhou, H., Zhou, W., Qi, W., Pu, J., & Li, H. (2021). Improving Sign Language Translation with Monolingual Data by Sign Back-Translation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2021)*, 1322-1332.
- Liang, Z., Li, H., & Chai, J., "Sign Language Translation: A Survey of Approaches and Techniques," *Electronics*, vol. 12, no. 12, article 2678, 2023, doi:10.3390/electronics12122678.
- Farooq, U., Mohd Rahim, M. S., Sabir, N., Hussain, A., & Abid, A., "Advances in Machine Translation for Sign Language: Approaches, Limitations, and Challenges," *Neural Computing and Applications*, vol. 33, pp. 14357-14399, 2021.

2.0 MANAGEMENT SUMMARY

2.1 Environment

2.1.1 Organizations Involved

[Identify the project sponsor, developer, user, and computer center or network in which the software will be implemented.]

Project Information

Process Owner	Alizah Basit
Process Owner	Amna Mansoor
Project Owner	Hafsa Salman
Project Type	in-house

Project Sponsor/ Director

No.	Name	K/C/D/I	Roles
1.	FAST NUCES, Karachi		Project Sponsor
2.	Mr. Basit Jasani		Project Director

Functional Team Members

No.	Name	K/C/D/I	Roles
1.	Alizah Basit		Functional Project Manager
2.	Amna Mansoor		Functional Team Member
3.	Hafsa Salman		Functional Team Member

Proposed Technical Team Members

No.	Name	K/C/D/I	Roles
1.	Alizah Basit		Technical Project Manager

2.	Amna Mansoor		Technical Team Member
3.	Hafsa Salman		Technical Team Member

2.1.2 Input/output

- **Input:** PSL gesture data (via camera), text, speech.
- **Output:** Translated text, speech, PSL animations/videos.

2.1.3 Processing

- Real-time video capture, gesture recognition using CNN + RNN/Transformers.
- NLP-based text-to-sign conversion.
- Backend processing with Python models connected to Flutter frontend.

2.1.4 Security

- User authentication (login/signup).
- Secure database (MongoDB) with encrypted credentials.
- Privacy compliance (no unnecessary storage of user camera input).

2.1.5 System Interaction

Independent Mobile App

2.1.6 Physical Environment

System environment or special conditions needed	
Estimated Number of Users	200-300
Internet or Intranet or both Usage	Both (online for translation, offline mode for dictionary/lessons).
Required storage? (require documents/files to be stored (upload feature) ?)	Yes (dictionary signs, user progress).
Database	Firebase
Application Server	Cloud
Special features? (<i>Please describe</i>)	Gamification

2.2 Current Functional Procedures

Currently, there is no automated PSL system. Non-hearing individuals rely on interpreters or manual teaching methods, which are limited and costly. No widely available PSL learning apps exist.

2.3 Functional Objectives

- Enable real-time PSL translation.
- Provide PSL learning resources in digital format.
- Track and assess user progress.
- Promote inclusivity in education and communication.

2.4 Assumptions and Constraints

- Assumes availability of PSL dataset.
- Limited computing resources for model training.
- The project must be completed within the academic year timeline.
- Budget limited to existing lab/software resources.

2.5 Risk

Risk Register				
Type of Risk *delete whichever not relevant	Risk Description	Mitigation Plan	Risk owner	Risk Timeline
Human Resource	<i>Team member unavailability</i>	<i>Tasks redistribution</i>	<i>Team Lead</i>	<i>Development Phase</i>
Business Process	Dataset Shortage	Use augmentation + open datasets	Developers	Initial Phase
Financial	Limited budget	Use open-source tools	Developers	Entire Project
Infrastructure/ Applications / Database	Hardware limits	Request resources from university	Developers	Mid-phase
Legal & Regulatory / Security & Privacy	User data misuse	Implement strict privacy policy	Developers	Development
Third Party Supplier & Outsourcing	API downtime	Offline fallback mode	Developers	Deployment

2.6 Methodology

- Literature review of ASL/PSL solutions.
- Dataset preparation and augmentation.

- Model selection: CNN + RNN/Transformers for gestures, NLP Transformers for text
- Development with iterative testing.

2.7 Statement of Needs (Business Case)

2.7.1

Strategic Planning Initiatives	(Please tick ✓)	Remarks
ATOP	☐	
ICTSP	☒	

2.7.2 Benefits and impact to the University

- Positions the university as a leader in inclusive and socially impactful research, addressing the needs of underrepresented communities.
- Provides a unique academic contribution by developing one of the first AI-based solutions for Pakistan Sign Language (PSL).

2.7.3 Please indicate the value added of the project to the University

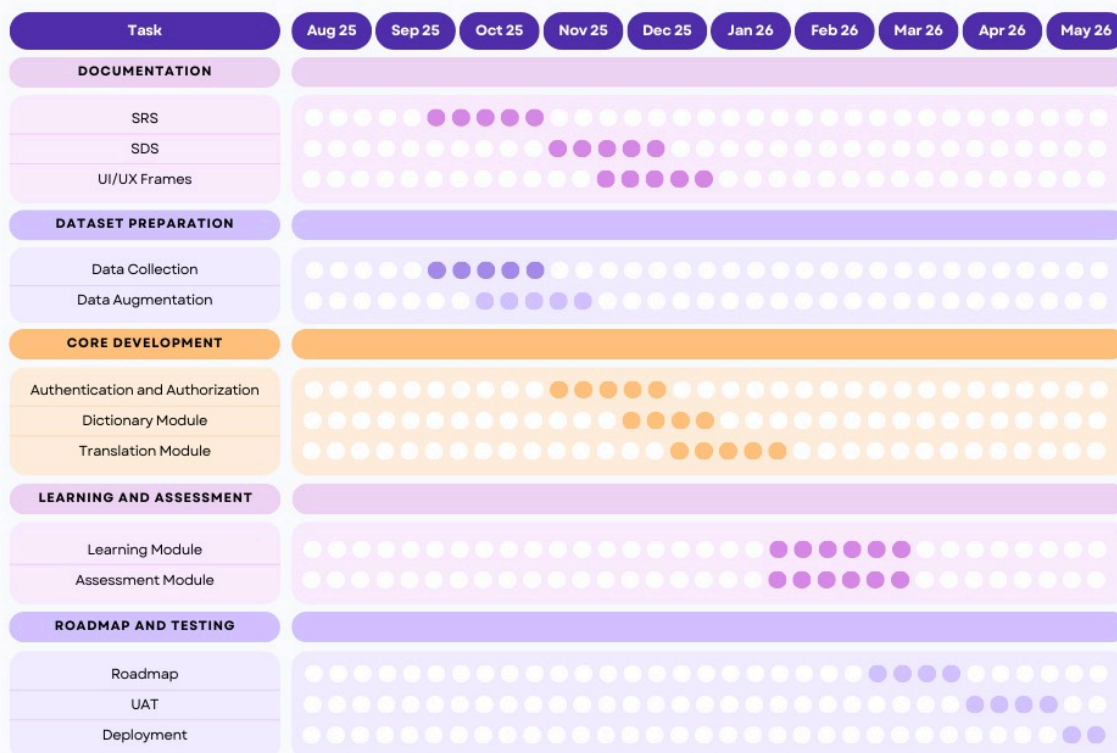
- Academic contribution (first PSL-focused app).
- Potential to expand into NGO based product
- ROI in terms of skill development and social impact.

3.0 AUTOMATION INITIATIVES

3.1 Time and Resource Costs Required

3.1.1 Project Timeline

Gantt Chart



No.	Task Name	Proposed Data
1.	Project Start	09/10/2025
2.	SRS	10/31/2025
3.	SDS	12/15/2025
4.	Authentication and Authorization Module	12/15/2026
5.	Dictionary Module	12/31/2025
6.	Translation Module	01/31/2026
7.	Learning Module	03/10/2026
8.	Assessment Module	03/10/2026
9.	Complete Solution Simulation and Testing	04/30/2026
10.	Deploy Solution	05/30/2026
11.	Project Complete	06/01/2026

Total Months Needed: 10

3.1.2 Project Cost

[State the estimated project cost and the value of the budget which has been secured. Please indicate the source of funds and attach the evidence that the budget has been secured as Appendix B: Source of Budget.]

Source of Budget		:
Bi l	Items	Budget Allocated
1.	Dataset prep	0 (open-source)
2.	Model Training	0 (university hardware)
3.	App development	0 (open-source frameworks)
GRAND TOTAL		Minimal (covered by university)

3.2 Existence of Similar Initiatives or System

No existing PSL-specific applications. ASL tools exist, but PSL is underserved.

3.3 Impacts

3.3.1 Equipment Impacts

- Standard mobile devices required.
- GPU-based system needed for training phase.

3.3.2 Software Impacts

- New mobile app development.
- No modification to existing university systems.

3.3.3 Organizational Impacts

- Students gain expertise in AI/ML and accessibility development.

3.3.4 Operational Impacts

- Users will learn PSL via mobile app.
- Improved accessibility in academic and social settings.

3.3.5 Developmental Impacts

- Requires dataset preparation.

- Security and privacy must be considered for user input.

3.3.6 Site or Facility Impacts

None (mobile-based system).

3.3.7 Security and Privacy Impacts

- Only essential user data stored.
- No sensitive biometric data retained.

3.4 Rationale for Recommendations

The proposed system is recommended because no similar PSL solution exists, it leverages modern AI technologies, and provides both translation and learning features in one platform. It directly addresses the communication gap and aligns with accessibility goals.